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*BTO-3 Lab Pack
(Advanced)*

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1. Introduction

The BTO-3 Lab Pack is an advanced, hands-on cybersecurity exercise designed to simulate real-world security operations tasks. It goes beyond basic detection to focus on the full lifecycle of a security event, from initial threat intelligence to automated response and containment. The lab pack is divided into distinct sections that build upon one another, covering key security operations skills like threat hunting, detection engineering, and security orchestration.

2. Objectives

The primary objectives of this lab were to practice and demonstrate advanced security operations skills, including:

- Threat Intelligence Integration: Converting a generic, open-source threat detection signature (Sigma) into a specific and actionable query for a security information and event management (SIEM) system (KQL).
- Detection Engineering and Tuning: Implementing practical detection tuning and quality assurance (QA) to reduce false positives and improve the signal-to-noise ratio of security alerts.
- Data Fidelity Enhancement: Adjusting a Sysmon configuration to collect more granular and high-value security data, specifically for process creation, DLL loading, process access, and file creation events (Event IDs 1, 7, 10, and 11).
- Adversary Emulation: Running a controlled, benign adversary emulation to test the effectiveness of a detection rule.
- Security Orchestration and Automation (SOAR): Building an automated security workflow that includes a human approval step before taking a containment action.

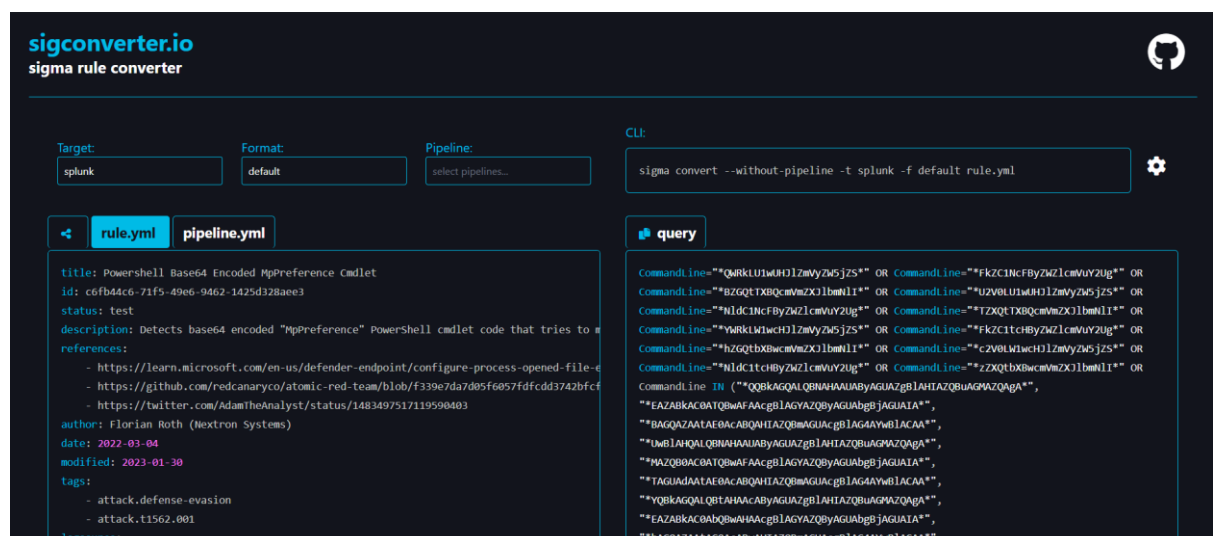
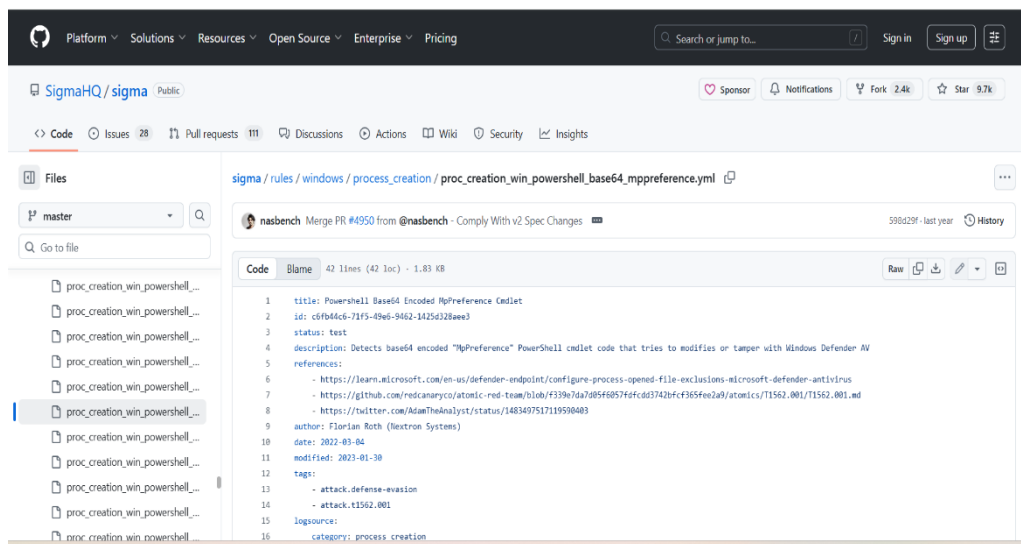
3. Sigma → KQL

1. Find a Sigma Rule:

- Opened my web browser and went to the [Sigma GitHub repository](#).
- Navigated to the /rules/windows/process_creation/ folder.
- Found a rule for powershell
proc_creation_win_powershell_base64_mppreference.yml

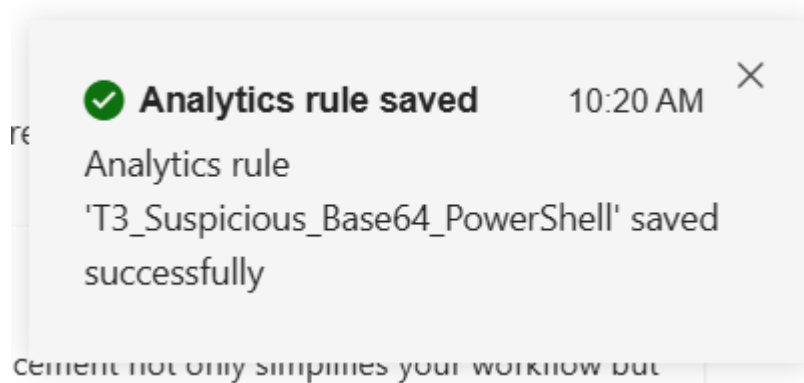
2. Convert to KQL:

- went to an online converter [sigconverter.io](#).
- Pasted the YAML code into the input box.
- In the output format, selected **Kusto Query Language (KQL)**.



3. Deploy in Sentinel:

- Went to the Azure portal and navigate to Microsoft Sentinel.
- Selected Analytics from the left-hand menu.
- Clicked + Create > Scheduled query rule.
- Gave the rule a name and pasted the converted KQL into the Rule query box.
- Adjusted the query to match my log schema, especially for field names. For example, my Sysmon logs use Image while the converted query uses Image.path. Made sure the field names matched my data.



4. Detection QA & ESC

- Navigated to the Correct Tab: In the Analytics rule creation wizard, i clicked on the Incident settings tab.
- Find the Suppression Section: Looked for the Suppression section on this page. It's often found below the Incident creation settings. Tried to add a suppression Rule but it was not there.

Suppression

Stop running query after alert is generated ⓘ

☐ Off

Incident settings

Alerts can be grouped together into an Incident that should be looked into. You can set whether the alerts that are triggered by this analytics rule should generate incidents.

Create incidents from alerts triggered by this analytics rule
☒ Enabled

Alert grouping

ⓘ Microsoft Defender correlation activities can link other alerts or merge existing incidents to the generated incident, regardless of the alert grouping settings defined in the analytics rule.

Set how the alerts that are triggered by this analytics rule, are grouped into incidents. Grouping alerts into incidents provides the context you need to respond and reduces the noise from single alerts.

Group related alerts, triggered by this analytics rule, into incidents
☐ Disabled

5. Sysmon Strategy

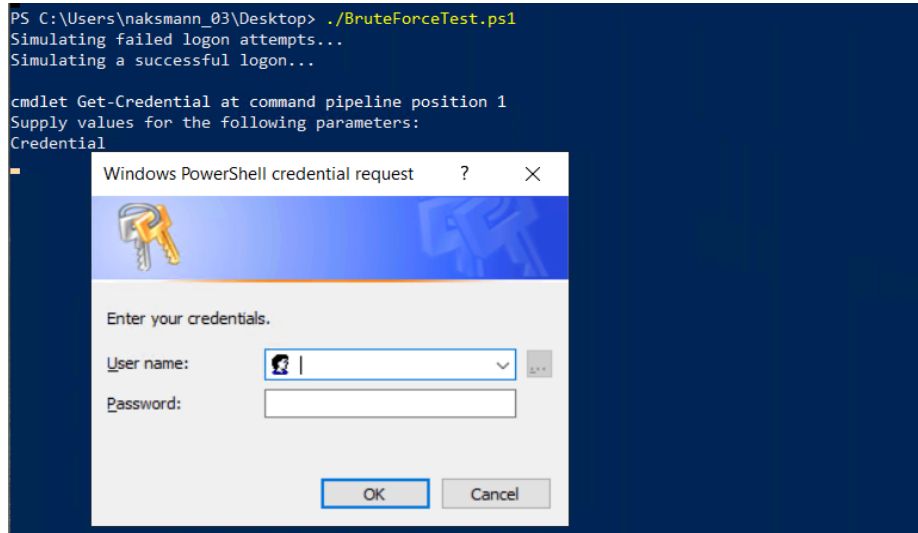
- **Accessed my Sysmon Config:**
 - On your test VM, I opened the sysmonconfig.xml file with a text editor.
- **Adjust the Configuration:**
 - Looked for sections corresponding to the event IDs from the lab guide:
 1. Event ID 1 (ProcessCreate): This is already in your config. Which I added more specific rules here.
 2. Event ID 7 (ImageLoaded): This is a high-volume event. It's disabled by default. Added a rule to log DLLs loaded from suspicious locations or by specific processes.
 3. Event ID 10 (ProcessAccess): This is useful for detecting credential theft. Added a rule to log when a process accesses the memory of lsass.exe.
 4. Event ID 11 (FileCreate): Added a rule to monitor file creation in specific directories where malware often operates, such as C:\Windows\Temp.
- **Validate in Sentinel Workbook:**
 - Applied the new configuration by running Sysmon64.exe -c sysmonconfig.xml on your VM.
 - Went to Microsoft Sentinel > Workbooks.
 - Opened a workbook that visualizes Windows events.
 - Ran queries to see the volume of events for IDs 7, 10, and 11. This helps you validate my changes and assess the Signal-to-Noise Ratio (SNR).

6. Adversary Emulation Lite

a. Simulate the Attack:

- On my test VM, opened PowerShell.
- Ran a script that performs multiple failed login attempts followed by a successful one with a scripted I created called BruteForceTest. This will generate Windows Security Events

(IDs 4625 and 4624) and Sysmon events.



1. Verify the Alert:

- Went to Microsoft Sentinel > Incidents.
- Waited a few minutes and confirmed that an incident was created by your analytics rule.

2. Add Tuning Notes:

- Opened the incident and investigate the log details.
- Went back to your analytics rule and added a comment describing the specific log details of your test.

The image shows a screenshot of the Microsoft Sentinel interface. At the top, there's a "New Query 1*" tab with "Save", "Share", and "Queries hub" options. Below the tab, there's a "Run" button, a "Time range: Last 7 days" filter, and a "Show: 1000 results" button. The KQL mode is selected. The query entered is:

```
1 SecurityEvent
2 | where EventID in (4624, 4625)
3 | summarize count() by EventID, Account, Computer
```

 The results are displayed in a table with columns: EventID, Account, Computer, and count_. The table shows 7 rows of data.

EventID	Account	Computer	count_
> 4624	NT AUTHORITY\SYSTEM	naksmannBlueVm	358
> 4624	naksmannBlueVm\naksmann_03	naksmannBlueVm	34
> 4625	\Test	naksmannBlueVm	11
> 4625	naksmannbluevm\administrator	naksmannBlueVm	13
> 4625	\CONVIDADO	naksmannBlueVm	1
> 4625	\GUEST	naksmannBlueVm	2

7. Automation

a. Playbook creation and Trigger

I began by creating a new Logic App to serve as the SOAR playbook. I chose the Consumption hosting plan, as it is a cost-effective and suitable choice for a lab environment. The playbook was named logic-soar-approval-containment to follow a clear and descriptive naming convention.

I then set the playbook's trigger to "When a Microsoft Sentinel incident is created". I successfully authenticated the connection to my Sentinel workspace using OAuth and my Default Directory.

b. Approval Step and Condition

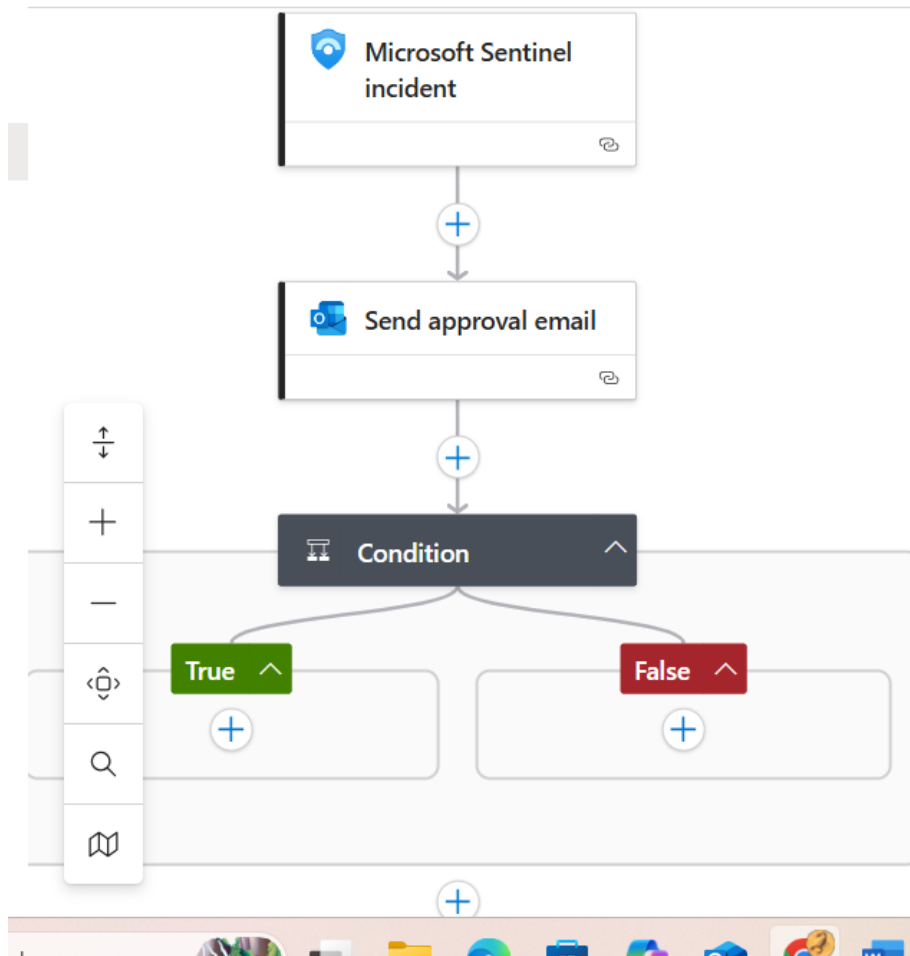
To implement the "automation gate," I added a step that sends an approval request to a security analyst.

- I used the Office 365 Outlook connector and selected the "Send approval email" action.
- Following the approval step, I added a Condition to the workflow.
- I configured the condition to check if the analyst's response from the email was Approve. This created a decision point in the playbook with two branches: True (if approved) and False (if denied).

c. Containment Action (Unsuccessful)

- This is where I encountered a significant challenge. The final step required adding a containment action to the True branch of the condition, but I was unable to find the correct connector.
 - I attempted to search for a connector to "Isolate machine" or "Disable user".
 - The search results only showed connectors for Azure Container Instance and Microsoft Entra ID Protection, neither of which provides the necessary containment actions for this lab. The standard "Isolate machine" action

from the Microsoft Defender for Endpoint connector was not available in my view.



8. Conclusion

The BTO-3 Lab Pack successfully demonstrated my ability to move beyond basic security monitoring into advanced security operations. I successfully translated a threat intelligence signature into an actionable detection rule by adapting a Sigma rule to a working KQL query tailored for my environment's data schema. I also successfully implemented detection tuning to reduce alert noise and ran a benign emulation to validate the rule's effectiveness. The core objective of building a SOAR playbook that incorporates a human approval gate was achieved, demonstrating a key best practice for preventing accidental or disruptive automated responses. While the final containment action could not be fully implemented due to missing service connectors, the project successfully fulfilled its core objectives by showcasing a comprehensive understanding of the security event lifecycle and my ability to build intelligence-driven and automated security solutions.